

A Voltage Drop-Based Approach to Include Cold Load Pickup in Design of Distribution Systems

Vinit Gupta, *Member, IEEE*, and Anil Pahwa, *Fellow, IEEE*

Abstract—In distribution systems with large residential load, thermostatically controlled devices contribute major portion of total load. Restoration of such systems after prolonged interruption may create a load significantly higher than the normal load. This condition is called cold load pickup (CLPU). The enduring component of CLPU may cause severe voltage drops along the feeders and transformer heating, thus making the system restoration difficult. This paper discusses an approach to include restoration under CLPU conditions in distribution system planning and expansion. The approach can be applied to determine the most optimal transformer size and feeder sizes while ensuring the minimal restoration time and acceptable voltages and transformer heating during CLPU conditions. The aim is to develop a tool that can be used in conjunction with the overall distribution system design to ensure quality power even in emergency conditions while minimizing the cost.

Index Terms—Cold load pickup, distribution system design, power system planning, power system restoration.

I. INTRODUCTION

DISTRIBUTION systems with large concentration of thermostatically controlled loads may result in loads significantly higher than the normal load during restoration after a prolonged outage due to cold load pickup [1], [2]. Of the different phases of cold load pickup, the first three which are in-rush, motor starting, and motor running have 5 to 15 times the normal current but they last less than 15 seconds. The fourth phase, which is the enduring phase, has load up to 5 times the normal load and it may last for several hours. Loss of diversity amongst thermostatically controlled loads is the main cause of the enduring phase of cold load pickup. The enduring phase can result in significant transformer overloading, conductor heating and unacceptable voltages at downstream service points in distribution systems during restoration. Therefore, the system must be designed so that it could be restored in least possible time without causing any voltage violations, feeder overload or unacceptable transformer heating.

Some researchers have attempted to include CLPU in distribution system design. Specifically, Ucak and Pahwa [2] investigated loading of power transformer and its hot-spot and top-oil temperatures during CLPU. They presented algorithms for calculating system restoration time, switching sequence and trans-

former temperatures for minimizing the customer interruption time under CLPU conditions. These algorithms are based on the transformer loading and temperature constraint specified in the IEEE guide for loading mineral-oil immersed transformers [3]. Later, Wakileh and Pahwa [4] expanded this work by discussing an optimization problem to select power transformer capacity and number of sectionalizing switches for a service area so as to avoid reaching overloading limits of transformer during cold load pickup. In essence, the previous research on this subject was a rigorous analysis of transformer overloading during cold load pickup but less attention was paid to overcome another severe problem, which is the excessive voltage drops on the feeders.

This paper presents an approach for distribution substation transformer and feeder system planning to ensure proper voltages, minimum customer interruption durations and acceptable transformer and feeder overload during CLPU conditions. The methodology discussed in this paper can be used in overall distribution system planning or expansion to examine the system reliability for emergency conditions such as CLPU.

II. DESIGN APPROACH

A distribution system is composed of many parts and the planner has to choose each part optimally while maintaining the overall economy of design [5]–[7]. The design approach discussed here addresses specific aspects of distribution systems related to conditions such as CLPU. Using this approach the planner can predict the most optimal transformer size, feeder sizes, location and number of sectionalizing switches and restoration time for the selected value of CLPU ratio (ratio of undiversified demand to diversified demand). Thus, the problem can be stated as:

$$\text{Minimize } f_{tc} = S_T f(S_T) + f_v + f_{ns} + f_{lol} + f_{cci} + f_{closs} + f_{tl} - f_{er} \quad (1)$$

where f_{tc} is the total cost in million \$, S_T is the transformer capacity in MVA, $f(S_T)$ is a function giving fixed cost of the transformer (\$/MVA), f_v is fixed cost of the feeder system, f_{ns} is cost of the sectionalizing switches, f_{lol} is present worth (PW) cost of the transformer's loss of life, f_{cci} is PW cost of customer interruptions, f_{closs} is PW cost of losses in the feeders, f_{tl} is PW cost of the transformer losses and f_{er} is PW of revenue generated due to faster restoration. Details of calculating some of these costs can be found in [8], [9]. Additional details are given in the Appendix.

The design constraints are:

- 1) During normal operating conditions voltage at any service point should be within $\pm 5\%$ of the system voltage and

Manuscript received May 9, 2003. This work was supported by the National Science Foundation under Grant 9721447.

V. Gupta was with Kansas State University, Manhattan, KS 66503 USA. He is now with Entergy Transmission, Pine Bluff, AR 71611 USA (e-mail: vgupta@entergy.com).

A. Pahwa is with Electrical and Computer Engineering Department, Kansas State University, Manhattan, KS 66503 USA (e-mail: pahwa@ksu.edu).

Digital Object Identifier 10.1109/TPWRS.2003.821440

during emergency conditions the service voltage could range between plus 6% to minus 8% (ANSI Range B) [10], [11].

- 2) Feeder conductors should not be loaded beyond 33% of their normal capacity [11].
- 3) Time of customer interruption should be minimized in case of sectional restoration.
- 4) Transformer loss of life and temperatures should be within IEEE limits [3].

A flow chart of the design approach including these constraints is shown in Fig. 1 and some of the important aspects are discussed in the following sections. K_p is the maximum load seen by transformer during CPLU divided by its rating.

A. CLPU Model

Several models ranging from piece-wise linear model to physically based models have been proposed for cold load pickup [2], [12]–[19]. This is only a selective list; a comprehensive list is available in [20]. This paper uses a delayed exponential model shown in Fig. 2. In this model the undiversified load S_U remains constant from R_i (restoration time) to t_i (the time when load diversity starts) and then load decreases exponentially to the diversified load S_D .

A mathematical expression for the total load of a section from the CLPU model shown in Fig. 2 is

$$S(t) = \left[S_D + (S_U - S_D)e^{-\alpha(t-t_i)} \right] u(t - t_i) + S_U [1 - u(t - t_i)] u(t - R_i) \quad (2)$$

where α is the rate of decay of load and $u(\cdot)$ is the unit step function.

B. Calculation of Voltage Drops

Typically electrical equipment are designed to operate within a narrow range of voltage around a standard utilization voltage. Utility standards are usually based on providing the customer service entrance voltage within $\pm 5\%$ of the nominal voltage (120 volts for North America). Since load during restoration can be very high due to the enduring component of CLPU, severe voltage drop can result and that can make system restoration difficult. To limit voltage drops, the feeders can be divided into several parts with sectionalizing switches and step-by-step restoration can be implemented.

During step-by-step restoration, loads on different sections of a feeder would be time dependent and at a given time would be at a different point on the CLPU model of Fig. 2. A procedure is developed to account for all the components of voltage drop created by these loads at various points on the feeder under different conditions. For illustration of the voltage drop calculations a feeder serving a triangular load area with uniform load density as shown in Fig. 3 is considered. Triangular service area is considered as an example, but the method can also be applied to other service area shapes. Also, in this paper the system is considered to be balanced and voltage drop calculations only for the main feeder are performed. However, an allowance of 3.4% maximum voltage drop on the laterals during emergency is included. Therefore, for a feeder with voltage 6% higher than rated at the substation, the total allowed voltage drop on the

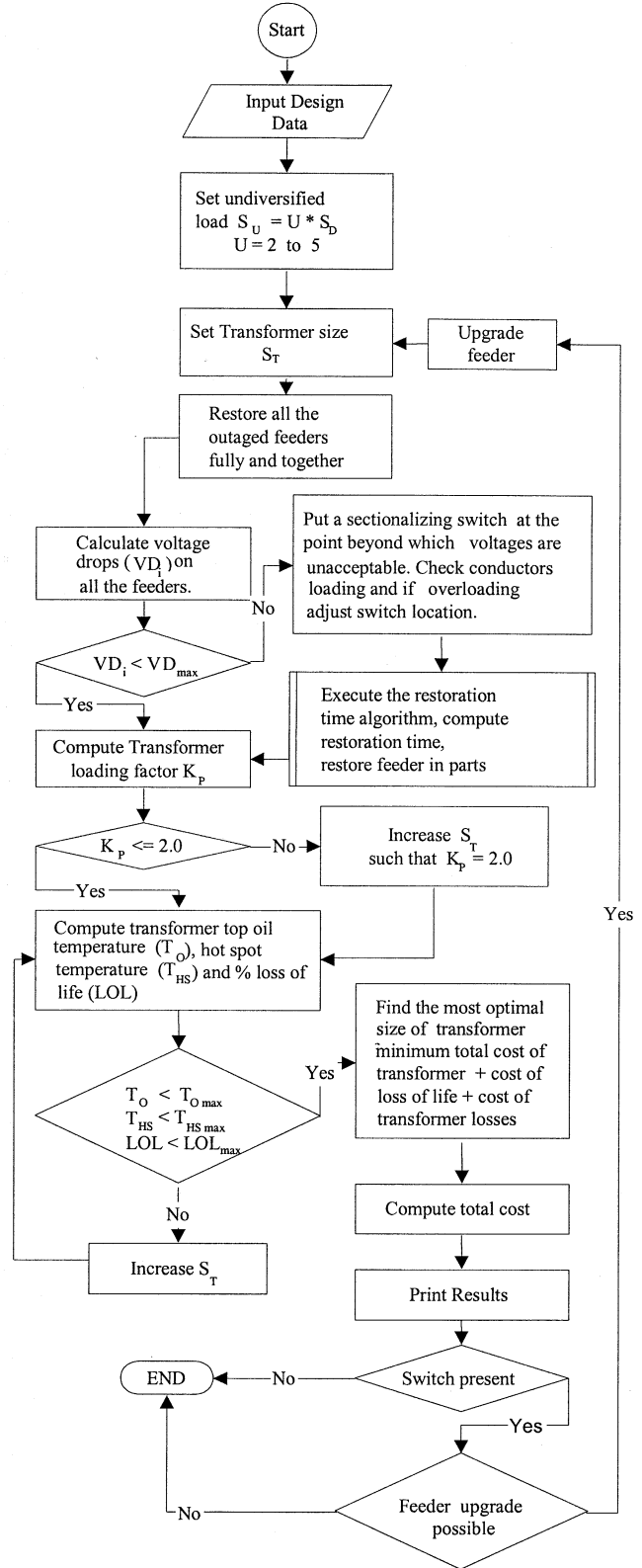


Fig. 1. Flowchart for the design algorithm.

main feeder would be 10.6% to maintain service voltage within the ANSI Range B. Unbalance and voltage drop on the laterals can be included in the model, but the problem would become a lot more complex and would require implementation of unbalanced system loadflow for voltage drop computations.

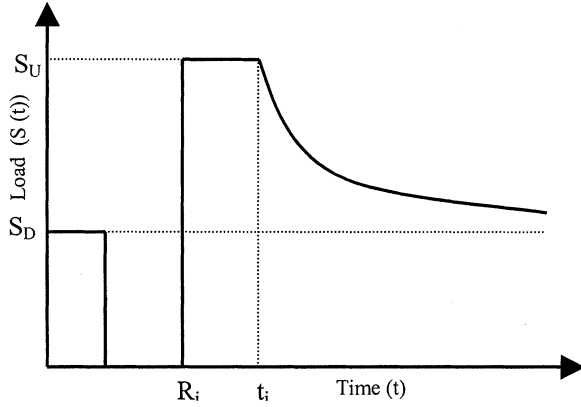


Fig. 2. Delayed exponential model of cold load pickup.

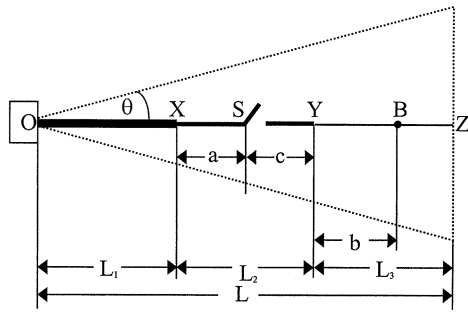


Fig. 3. Triangular service area with a sectionalizing switch.

The feeder shown in Fig. 3 has only one sectionalizing switch, but the method can be easily generalized to any number of switches. It is also assumed that the main feeder is tapered and it consists of a maximum of three different conductors. Other variables considered in the calculation are

- Length of the feeder='L' miles;
- Total area served by the feeder='A' square miles;
- Load density='M' MVA/square miles.

It is assumed that the main breaker at the substation is closed at $t = 0$ to restore power to loads connected to the feeder between points O and S, and at time t_1 the sectionalizing switch is closed to supply power to the rest of the feeder. Hence, the voltage drop up to 'B' at time $(t_1 + \epsilon)$, where ϵ is a very small number, is given by

$$VD_{OB} = VD_{OX} + VD_{XS} + VD_{SY} + VD_{YB} \quad (3)$$

Each of these terms is composed of various parts. For example, VD_{OX} is composed of two parts, which are voltage drop due to the distributed load between O and X, and voltage drop due to the lumped load of the area between X to Z. Thus

$$VD_{OXDistributed} = \left(\frac{2}{3}\right) S_{3-\phi} \left(\frac{M_{11}}{M}\right) K_1 \left(\frac{L_1^3}{L^2}\right) \quad (4)$$

$$\text{and } VD_{OXLumped} = (A_{XS}M_{11} + A_{SZ}M_U)K_1L_1 \quad (5)$$

where $S_{3-\phi}$ is the total 3-phase load served in MVA, M_{11} is the load density of section OS at time $t_1 + \epsilon$ (calculated using (2)), K_1 is the voltage drop constant [5] for conductor OX, A_{XS} is the area between points X and S, A_{SZ} is the area between points S

and Z, and M_U is the undiversified load density. Derivation of (4) is based on considering a differential strip between O and X and then integrating over the length OX [5], [10].

Similar expressions obtained for other sections to compute the voltage drop at B are

$$VD_{XS} = \left(\frac{S_{3-\phi}}{L^2}\right) \left(\frac{M_{11}}{M}\right) K_2 a^2 \left(L_1 + \left(\frac{2}{3}\right) a\right) + (A_{SZ}M_U)K_2 a \quad (6)$$

$$VD_{SY} = \left(\frac{S_{3-\phi}}{L^2}\right) \left(\frac{M_U}{M}\right) K_2 c^2 \left(L_1 + a + \left(\frac{2}{3}\right) c\right) + (A_{YZ}M_U)K_2 c \quad (7)$$

$$VD_{YB} = \left(\frac{S_{3-\phi}}{L^2}\right) \left(\frac{M_U}{M}\right) K_3 b^2 \left(L_1 + L_2 + \left(\frac{2}{3}\right) b\right) + (A_{BZ}M_U)K_3 b \quad (8)$$

where K_2 and K_3 are the voltage drop factors for conductors used for sections XY and YZ, and a , b , and c are distances shown in Fig. 3. Using this approach voltage drops can be calculated at any point along the feeder and at any time.

C. Calculation of Restoration Time

If it is found that the system cannot be restored fully in one step due to the voltage drop constraint, we need to find location and restoration time of the sectionalizing switches. The point on the feeder beyond which voltage drop is unacceptable decides location of the first switch. In some cases with very high CLPU ratio, it may not be possible to restore the system with a single sectionalizing switch. In such cases, we need to locate another sectionalizing switch and find its restoration time such that the total restoration time is minimized and no conductor is overloaded. An algorithm for deciding location of the second switch and restoration time of both switches is outlined as follows:

- 1) Set initial restoration time t_{s1} of the first switch at the mid point of t_i and an approximate maximum time t_{max} at which the load is expected to gain full diversity (see Fig. 1). Calculate voltage drop at the end of feeder at t_{s1} .
- 2) If voltage drop is more than the limit, restoration time will be between t_{s1} and t_{max} else it will be between t_i and t_{s1} . Apply binary search in the appropriate region to find restoration time of the first sectionalizing switch such that voltage drop at the end of the feeder upon closing the sectionalizing switch is just equal to the limiting value.
- 3) If no feasible restoration time can be found, the feeder needs another sectionalizing switch.
- 4) Reset restoration time of the first sectionalizing switch t_{s1} to $t_i + h$, where h is a small increment in time.
- 5) Place the second switch at the point beyond which voltage drop is unacceptable. Find its restoration time t_{s2} with a binary search such that no voltage drop constraints are violated.
- 6) Increment restoration time of the first sectionalizing switch by h and repeat step 5 until t_{s1} is equal to t_{max} .
- 7) Choose values of t_{s1} , t_{s2} and the second switch location corresponding to the minimum t_{s2} .

Loading of conductors at the instant of restoring each sectionalizing switch must be examined and if overloading occurs, the

sectionalizing switch must be moved closer to the substation or the conductor size must be increased.

D. Optimizing the Transformer Size

Starting value of the transformer size is selected such that it would be loaded within a range from 80% to 90% of the rating under normal operation (diversified conditions). However, the design procedure is general enough to allow any starting value of the transformer size. The design algorithm selects the best size of transformer for the given conditions through a search process while keeping the transformer temperatures and loss of life within limits.

The smallest size of transformer needed for the system is that for which the winding hot-spot temperature, top-oil temperature, loss of life and loading ratio (K_P) are within limits during restoration. The cost associated with the transformer consists of three components; the first is the capital cost of transformer, the second is the cost of loss of life, and the third is the cost of losses. The cost of loss of life and the cost of losses decrease with an increase in the transformer size, but the capital cost increases with bigger size. To find an optimal value, the size of transformer can be increased in steps and the total cost (summation of the cost of transformer, present worth cost of loss of life, and present worth cost of losses) over a specified period can be computed. The cost of loss of life is computed using the aging acceleration factors given in [3] for a specified number of CLPU events during this period, for example once in 5 years. The minimum value of cost decides the rating of transformer. Transformer sizes in increments of 1 MVA were considered for the results given in this paper. However, the algorithm can be easily modified to consider a selected set of typical transformer sizes.

E. Upgrading the Feeder System

After computing the total cost with a given set of conductors for the feeder, benefits of using bigger conductors were examined. Using bigger conductors would reduce voltage drop and the system can be restored faster. Although the cost of installation and maintenance is higher for the bigger conductors, reduction in the customer interruption duration could be quite significant. Hence, repeating the design process with different conductor sets allows selection of the most suitable conductor set for the cold load pickup conditions.

III. CASE STUDY

The methodology discussed in this paper is applied to design a sample distribution system. The parameters of this system from [11] are

- subtransmission system voltage: 138 kV;
- feeder system voltage: 12.47 kV;
- substation spacing: 4.56 mi;
- area served by each feeder: 2 sq-mi;
- number of feeders per transformer: 3;
- load density: 3.61 MVA/sq-mi;
- load power factor: 0.9;
- loss factor (F_{LS}): 0.48;
- study duration in years (N): 30;

TABLE I
ECONOMIC LOADING RANGES AND LENGTHS OF CONDUCTORS

Conductor	Economic loading range (MW)	Length (miles)
#2	0 – 1.16	0.319
4/0	1.16 – 4.0	0.971
636 kcmil	4.0 – 12.0	2.107

TABLE II
RESULTS OF DIFFERENT CASES WITH CLPU RATIO OF 4

	636kcmil, 4/0, and #2	795 kcmil, 4/0, and #2 with length adjustment	795 kcmil, 4/0, and #2 without length adjustment
Transformer Capacity (S_T), MVA:	30	29	32
Maximum transformer loading ratio K_P :	1.93	1.93	1.91
Transformer loss of life, %:	0.0570	0.0568	0.0450
Location of first sectionalizing switch (distance from substation), miles:	1.86	1.83	1.91
Restoration time of first sectionalizing switch, minutes:	30	34	24
Location of second sectionalizing switch, miles:	2.12	2.12	2.12
Restoration time of second sectionalizing switch, minutes	59.29	65.25	48.60
Total Cost, million \$:	2.419	2.436	2.473

- frequency of CLPU event: 0.2/yr;
- discount rate in % (r): 12.

The first step in design is to choose the most economical conductors for the given loading under normal conditions as discussed in [9] and [11]. The direct length of each feeder based on the substation spacing would be 2.28 miles, but due to routing the actual length can be higher than the direct length. A factor of 1.5 (ratio of the feeder length to the direct distance from the substation to the end of feeder) is used to compensate the length of feeder for routing, which gives a total length of 3.4 miles for the feeder. Lengths of conductors and their economic loading for a specific conductor set for the feeders can then be computed for a fixed maximum reach of 3.4 miles at the normal peak loading [9], [11], which are given in Table I for a selected set of 636 kcmil, 4/0 and #2 conductors. The smallest conductor (#2) was de-rated slightly (calculated range was 0.14 MW) to avoid overload at very high CLPU ratios. Results for this set of conductors for a CLPU ratio (U) of 4 are shown in Table II.

The transformer capacity of 30 MVA is computed based on the worst case of all three feeders on outage and thus being restored simultaneously. Note that with this rating the transformer would be loaded to 72.2% of its rating under the peak conditions during normal operation (peak load under normal operation is computed based on load density of 3.61 MVA/sq-mile and 6 sq-miles service area for the transformer). This loading

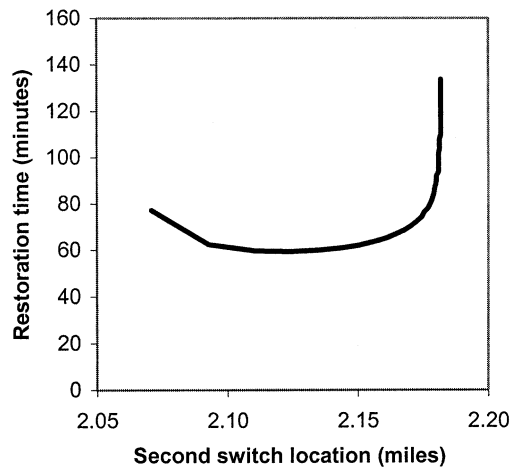


Fig. 4. Effect of location of the second sectionalizing switch on restoration time.

TABLE III
ECONOMIC LOADING RANGE AND LENGTHS OF CONDUCTORS WITH THE LARGEST CONDUCTOR UPGRADED TO A BIGGER SIZE

Conductor	Economic loading range (MW)	Length (miles)
#2	0 – 1.16	0.319
4/0	1.16 – 4.5	1.195
795 kmil	4.5 – 13.6	1.883

level is consistent with the loading levels selected using normal design procedures. Fig. 4 shows graph of the second switch location with respect to the total restoration time. The restoration time decreases initially and remains almost constant and then starts increasing rapidly. Thus, a second switch can be placed anywhere between 2.12 miles to 2.14 miles from the substation without much difference in restoration time.

Next, upgrading the biggest conductor from 636 kmil to 795 kmil was considered. Lengths and economic loading for these conductors were recomputed and are shown in Table III. The results for this case are given in column three of Table II. The restoration time is 65.25 minutes with a total cost of \$2.436 million for this case. The transformer temperatures and loading were within IEEE limits and the second switch decision curve was similar to the previous case. It is obvious that upgrading 636 kmil conductor to 795 kmil did not reduce the system restoration time, but the total cost increased. Further examination revealed that adjustment of conductor lengths based on economic loading ranges resulted in this increase in the restoration time, that is, when 636 kmil conductor was upgraded to 795 kmil, the length of the largest conductor was reduced to 1.883 miles while the length of 4/0 was increased to 1.195 miles. Thus, increase in the length of 4/0 nullified the effect of the bigger conductor and rather produced more voltage drops along the feeder. This problem can be corrected by not adjusting the lengths of conductors during upgrade. Hence, with 2.107 miles of 795 kmil, 0.971 miles of 4/0 and 0.319 miles of #2, the restoration time reduced to 48.6 minutes with the total cost of \$2.473 million. Complete results for this case are given in the last column of Table II.

TABLE IV
RESULTS OF FEEDER UPGRADE FOR DIFFERENT CLPU RATIOS

CPLU Ratio	636 kmil, 4/0, #2 Conductors		795 kmil, 4/0, #2 Conductors	
	Cost (\$ M)	Restoration Time (minutes)	Cost (\$ M)	Restoration Time (minutes)
4.5	2.429	94.09	2.482	65.00
4.0	2.419	59.29	2.473	48.60
3.0	2.362	32.92	2.436	25.95
2.0	2.358	15.58	2.401	00.00

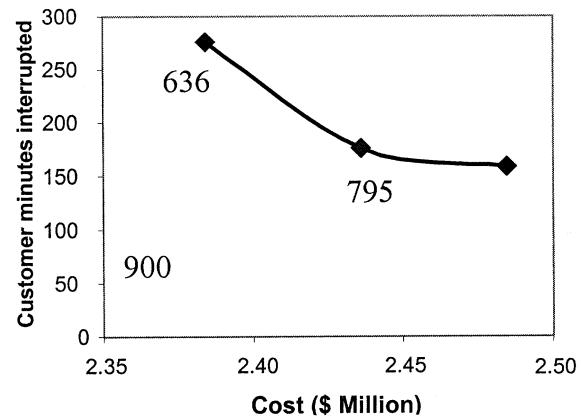


Fig. 5. Total customer minutes interrupted for $U = 4.0$ for different sizes of the largest conductor.

A comparison of the second and the fourth column of Table II shows that an increase in the total cost of \$0.054 million resulted in reduction of the restoration time by 10.69 minutes. In addition to upgrade of feeders, the transformer size increased from 30 to 32 MVA to accommodate the extra load due to faster restoration. Several other cases with different combinations of conductor sets and CLPU ratios were tried. Lengths of the feeders were not adjusted with change in conductor size. Selected results for these cases are shown Table IV. It is evident that these results are consistent with that of Table II. However, the reduction in the restoration time due to feeder upgrade for CLPU ratio of 4.5 is 29.09 minutes, which is significantly higher than 10.69 minutes obtained for CLPU ratio of 4. Also, for a CLPU ratio of 2, the need for a sectionalizing switch disappeared and the restoration time reduced to zero.

Further upgrade of the largest conductor from 795 kmil to 900 kmil for CLPU ratio 4 reduced the restoration time to 46.9 minutes for a total cost of \$2.522 million or the restoration time reduced by an additional 1.7 minutes for an increase in cost of \$0.049 million. Thus this upgrade does not seem to provide much benefit. To investigate this issue further, total customer minutes interrupted for the three conductor sets for CLPU ratio of 4 were determined. The total customer minutes interrupted during CLPU are important since they contribute directly to the reliability index SAIDI (System Average Interruption Duration Index). Total customer minutes interrupted were computed by assuming the average load of a household to be approximately 8 kW. Fig. 5 shows a plot of total customers minutes interrupted

versus cost for the three conductor sets. The total customer minutes interrupted fall rapidly when 636 kcmil is upgraded to 795 kcmil, but decrease is not significant with further upgrade to 900 kcmil. Hence, the best solution for this system is to upgrade the conductor to 795 kcmil to obtain the most value from investment.

IV. CONCLUSION

In this paper, a procedure to design distribution systems to accommodate cold load pickup is presented. Results show that lengths of the feeders should not be adjusted while upgrading the conductors to larger sizes. The incremental cost of upgrade must be compared with reduction in the total customer minutes interrupted to make the best decision on selecting size of conductors for the system.

The size of the transformer obtained with the method presented in this paper is to accommodate cold load pickup. Other considerations may be included in the design process to determine appropriate size of transformer. In the system under consideration, voltage drop on the feeders determined locations of sectionalizing switches while transformer loading was not a concern. In general, in semi-urban and rural systems, which are voltage constrained, voltage drop on feeders will play a more important role in selecting location of sectionalizing switches. In high-density urban systems, which have short feeders, transformer size would decide locations of the sectionalizing switches since voltage drop would not be very high.

The design approach has been presented with a specific case of triangular service area with a main feeder. However, this method can be applied to other shapes of service areas and feeder design during the planning stage. It can also be used on the existing systems to study the effect of possible CLPU conditions. The approach is also quite effective for including CLPU in the conductor inventory selection or it could be used during a feeder upgrade.

APPENDIX I

COST OF CUSTOMER INTERRUPTION (CCI)

Based on the data reported in [21], an equation to fit the data related to the residential load is obtained, which is

$$f(\text{CCI}) = a + bx + cx^2 + dx^3 \quad \frac{\$}{\text{kW}}$$

with $a = -0.001692$, $b = 0.0033217$, $c = 8.1879 \times 10^{-5}$, $d = -4.309389 \times 10^{-8}$, and x is the interruption duration in minutes. Thus, the present worth of cost of customer interruption for the study duration is given by

$$f_{\text{cci}} = \text{PW}_{\text{CCI}} f(\text{CCI})(\text{kW}_{\text{ns}}) \frac{1}{N_{\text{clpu}}}$$

where PW_{CCI} is the present worth factor for calculating cost of customer interruption over the total study period, kW_{ns} is the power not served in kW, and N_{clpu} is the time interval in years during which one CLPU event is expected to occur.

APPENDIX II

COST OF LOSSES IN FEEDERS

Consider a strip of width dx at distance x from point O (Fig. 3). The load flowing through the strip is equal to the load contained in the trapezoidal area formed between the strip and point Z.

$$\text{Area of trapezoid} = L^2 \tan \theta - x^2 \tan \theta$$

$$\text{Load (3-phase) through the strip} = (L^2 - x^2) \tan \theta M \cos \varphi \text{ MW.}$$

Current I (per phase) flowing through the strip

$$= \frac{(L^2 - x^2) \tan \theta M \times 10^3}{\sqrt{3}V} \text{ A.}$$

With $C = ((\tan \theta M \times 10^3)/\sqrt{3}V)$, $I = C(L^2 - x^2) \text{ A.}$

Per phase average losses in the strip $= I^2 F_{\text{LS}} R_1 dx$

$$= C^2(L^2 - x^2)^2 F_{\text{LS}} R_1 dx \text{ Watts}$$

where F_{LS} is the loss factor.

Therefore, losses (kWh per year)

$$= 8.760 C^2 (L^2 - x^2)^2 F_{\text{LS}} R_1 dx$$

Total three phase losses (kWh per year) in conductor OX are

$$\begin{aligned} \text{Loss}_{\text{OX}} &= 3 \times 8.760 C^2 F_{\text{LS}} R_1 \int_0^{L_1} (L^2 - x^2)^2 dx \\ &= 3 \times 8.760 C^2 F_{\text{LS}} R_1 \left(L^4 L_1 + \frac{L_1^5}{5} - \frac{2L^2 L_1^3}{3} \right) \end{aligned}$$

Similarly, expressions for losses (kWh per year) in conductor sections 'XY' and 'YZ' are obtained, which are

$$\begin{aligned} \text{Loss}_{\text{XY}} &= 3 \times 8.760 C^2 F_{\text{LS}} R_2 \\ &\times \left[\left(L^4 l_2 + \frac{l_2^5}{5} - \frac{2L^2 l_2^3}{3} \right) - \left(L^4 L_1 + \frac{L_1^5}{5} - \frac{2L^2 L_1^3}{3} \right) \right] \end{aligned}$$

$$\begin{aligned} \text{Loss}_{\text{YZ}} &= 3 \times 8.760 C^2 F_{\text{LS}} R_3 \\ &\times \left[\left(\frac{8}{15} L^5 \right) - \left(L^4 l_2 + \frac{l_2^5}{5} - \frac{2L^2 l_2^3}{3} \right) \right] \end{aligned}$$

where $l_2 = L_1 + L_2$.

Thus, the present worth cost of losses on the feeders for the selected configuration are given by,

$$f_{\text{loss}} = (\text{Loss}_{\text{OX}} + \text{Loss}_{\text{XY}} + \text{Loss}_{\text{YZ}}) D \text{PW}_L$$

where D is the cost of energy in \$/kWh and PW_L is the present worth factor used for loss calculation.

REFERENCES

- [1] J. E. McDonald, A. M. Bruning, and W. R. Mahieu, "Cold load pickup," *IEEE Trans. Power App. Syst.*, vol. 98, pp. 1384–1386, July/Aug. 1979.
- [2] C. Uçak and A. Pahwa, "An analytical approach for step-by-step restoration of distribution systems following extended outages," *IEEE Trans. Power Delivery*, vol. 9, pp. 1717–1723, July 1994.
- [3] *IEEE Guide for Loading Mineral Oil-Immersed Transformers ANSI/IEEE C57.91-1995*, IEEE, Piscataway, NJ, 1995.

- [4] J. J. Wakileh and A. Pahwa, "Optimization of distribution system design to accommodate cold load pickup," *IEEE Trans. Power Delivery*, vol. 12, pp. 339–345, Jan. 1997.
 - [5] T. Gonen, *Electric Power Distribution System Engineering*. New York: McGraw-Hill, 1984.
 - [6] T. M. Taylor, H. L. Willis, and M. V. Engel, "New considerations for distribution network planning and design," in *Proc. Elect. Dist. Part 1: Contributions. CIRED. 14th Int. Conf. Exhibition IEE*, vol. 6, 1997, Conf. Pub. no. 438, pp. 1/1–1/5.
 - [7] S. K. Khator and L. C. Leung, "Power distribution planning: a review of models and issues," *IEEE Trans. Power Syst.*, vol. 12, pp. 1151–1159, Aug. 1997.
 - [8] J. J. Wakileh and A. Pahwa, "Distribution system design optimization for cold load pickup," *IEEE Trans. Power Syst.*, vol. 11, pp. 1879–1884, Nov. 1996.
 - [9] S. Mandal and A. Pahwa, "Optimal selection of conductors for distribution feeders," *IEEE Trans. Power Syst.*, vol. 17, pp. 192–197, Feb. 2002.
 - [10] *Electric Utility Engineering Reference Book Vol. 3: Distribution Systems*. East Pittsburgh, PA: Westinghouse Electric Corp., 1959.
 - [11] H. L. Willis, *Power Distribution Planning Reference Book*. New York: Marcel Dekker, 1997.
 - [12] R. L. Wilde, "Effects of cold load pickup at the distribution substation transformer," *IEEE Trans. Power App. Syst.*, vol. PAS-104, pp. 704–710, Mar. 1985.
 - [13] J. Aubin, R. Bergeron, and R. Morin, "Distribution transformer overloading capability under cold-load pickup conditions," *IEEE Trans. Power Delivery*, vol. 5, pp. 1883–1891, Nov. 1990.
 - [14] S. Ihara and F. C. Schweppe, "Physically based modeling of cold load pickup," *IEEE Trans. Power App. Syst.*, vol. PAS-100, pp. 4142–4150, Sept. 1981.
 - [15] W. W. Lang, M. D. Anderson, and D. R. Fannin, "An analytical method for quantifying the electrical space heating component of a cold load pick up," *IEEE Trans. Power App. Syst.*, vol. PAS-101, pp. 924–932, Apr. 1982.
 - [16] C. Y. Chong and R. P. Malhamé, "Statistical synthesis of physically based load models with applications to cold load pickup," *IEEE Trans. Power App. Syst.*, vol. PAS-103, pp. 1612–1628, July 1984.
 - [17] A. Pahwa and C. W. Brice, "Modeling and system identification of residential air conditioning load," *IEEE Trans. Power App. Syst.*, vol. PAS-104, pp. 1418–1425, June 1985.
 - [18] D. Athow and J. Law, "Development and applications of a random variable model for cold load pickup," *IEEE Trans. Power Delivery*, vol. 9, pp. 1647–1653, July 1994.
 - [19] J. Law, D. Minford, L. Elliott, and M. Storms, "Measured and predicted cold load pick up and feeder parameter determination using the harmonic model algorithm," *IEEE Trans. Power Syst.*, vol. 10, pp. 1756–1764, Nov. 1995.
 - [20] C. Uçak, "Restoration of distribution systems following extended outages," Ph.D. dissertation, Kansas State University, Manhattan, KS, 1994.
 - [21] R. Billington and G. Wacker, "Customer cost of electric service interruptions," *Proc. IEEE*, vol. 77, pp. 919–930, June 1989.
- Vinit Gupta** (M'03) received the B.E. in electrical engineering from the University of Roorkee, Roorkee, India, in 1998 and the M.S. degree in electrical engineering from Kansas State University, Manhattan, in 2002. Currently, he is an EMS Engineer at Entergy Transmission, Pine Bluff, AR.
- Anil Pahwa** (F'03) received the B.E. (hons.) degree in electrical engineering from Birla Institute of Technology and Science, Pilani, India, in 1975, the M.S. degree in electrical engineering from the University of Maine, Orono, in 1979, and the Ph.D. in electrical engineering from Texas A&M University, College Station, in 1983. Currently, he is Professor in the Electrical and Computer Engineering Department at Kansas State University, Manhattan, where he has been since 1983. He worked at ABB-ETI, Raleigh, NC, during sabbatical from 1999 to 2000. His research interests include distribution automation, distribution system planning and analysis, and intelligent computational methods for power systems analysis.